

Mansur Maturidi Arief

Executive Director, Center for AI Safety, Stanford University

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EDUCATION

- Carnegie Mellon University, Pittsburgh PA, USA 2018–2023
PhD in Mechanical Engineering
- University of Michigan, Ann Arbor MI, USA 2016–2018
MSE in Industrial and Operations Engineering
- Sepuluh Nopember Institute of Technology (ITS), Surabaya, Indonesia 2010–2014
BE in Industrial and Systems Engineering

EMPLOYMENT

- Stanford University, Stanford, CA, USA 2024–current
Executive Director, Center for AI Safety
Research Engineer, Stanford Intelligent Systems Lab (SISL) & Mineral-X
- Stanford University, Stanford, CA, USA 2023–2024
Postdoctoral Research Scholar, Stanford Intelligent Systems Lab (SISL)
- Robert Bosch LLC (Bosch), Pittsburgh, PA, USA May–August 2021
Machine Learning/Computer Vision Research Intern
- University of Michigan, Ann Arbor, MI, USA 2017–2018
Research Assistant, University of Michigan Transportation Research Institute (UMTRI)
- Sepuluh Nopember Institute of Technology, Surabaya, Indonesia 2012–2016
Associate, Office of International Affairs
Research Assistant, Logistics and Supply Chain Management (LSCM) Laboratory
Research Assistant, Center for Transportation Research

TEACHING EXPERIENCE

- Analytics Projects (*Remote*), Co-Instructor, SIMT ITS 2024–2025
- Trustworthy AI Autonomy (24784), Teaching Assistant, CMU 2021
- Intro to Scientific Computing (24781), Teaching Assistant, CMU 2020
- Supply Chain Management, Teaching Assistant, IndonesiaX 2016
- Network Design and Distribution Management, Teaching Assistant, ITS 2014
- Industrial Systems Design, Teaching Assistant, ITS 2013

AWARDS & CREDENTIALS

- Best Overall Presentation, IMMAN 2024 2024
- Outstanding Indonesian Diaspora 2024, Consulate General of the Republic of Indonesia in San Francisco 2024
- Cyber-Physical System (CPS) Rising Stars, LinkLab, University of Virginia 2024
- Early Career Mini Research Grant, Indonesian Association of International Scientists (I-4) 2024
- Best Poster, Michigan Student Symposium on Interdisciplinary Statistical Sciences (MSSISS) 2018
- Best Paper, Power and Energy Symposium, International Conference on Science and Technology (ICST) 2018
- LPDP Scholarship Awardee, Indonesian Endowment Fund for Education 2016
- 1st Place, National Industrial Engineering Challenge, University of Andalas 2014

PUBLICATIONS

- **Arief, Mansur**, Yasmine Alonso, CJ Oshiro, William Xu, Anthony Corso, David Zhen Yin, Jef Caers, and Mykel Kochenderfer. Managing Geological Uncertainty in Critical Mineral Supply Chains: A POMDP Approach with Application to US Lithium Resources. *arXiv preprint, under review*, 2025.
- Rusman, Muhammad, Andry Yusdi, Dwi Handayani, **Mansur Arief**. “Financial and Service Optimization for Distressed Bus Rapid Transit Systems Using Goal Programming in Emerging Economies”. *Under review*, 2025.
- Kinnarkar, Riya, **Mansur Arief**. “Optimized Renewable Energy Planning Model for Socially-Equitable Electricity Coverage in the US”. *arXiv preprint, under review*, 2025.
- Sharma, Armaan, **Mansur Arief**. “Hydrogen in Aviation: Evaluating the Feasibility and Benefits of a Green Fuel Alternative.” *Journal of High School Research*, 2(1), 2025.
- **Arief, Mansur**, Yuanlu Bai, Shengyi He, Wenhao Ding, Zhiyuan Huang, Henry Lam, Bo Li, Ding Zhao. “Certifiable Deep Importance Sampling for Rare-Event Simulation of Black-Box Systems.” *To Appear in Operations Research (INFORMS)*.
- Lange, Bernard, Anil Yildiz, **Mansur Arief**, Shehryar Khattak, Mykel Kochenderfer, Georgios Georgakis. "General-Purpose Robotic Navigation via LVLM-Orchestrated Perception, Reasoning, and Acting." *arXiv preprint, under review*.
- **Arief, Mansur**, Zhiyuan Huang, Suntichai Kotcharin, Ivan Kristianto Singgih. “Artificial Intelligence, Cyber-Physical Systems, and Decision Making in Operations and Supply Chain Management.” *Operations and Supply Chain Management: An International Journal*, 18(2). OSCM Forum, 2025.
- **Arief, Mansur**, Mike Timmerman, Jiachen Li, David Isele, Mykel Kochenderfer. “Importance Sampling Guided Meta Training for Intelligent Agents under Highly Interactive Driving Scenarios.” *IEEE Robotics Automation Letters*. IEEE, 2025.
- Kruse, Liam, Alexandros Tzikas, Harrison Delecki, **Mansur Arief**, Mykel Kochenderfer. “Enhanced Importance Sampling through Latent Space Exploration in Normalizing Flows.” *Proceedings of the 39th Annual AAAI Conference on Artificial Intelligence (AAAI-25)*. AAAI, 2025.
- Ward, Isaac, Dylan M. Asmar, **Mansur Arief**, Mykel Kochenderfer, Jana Krystofova Mike. “Optimal Control of Mechanical Ventilators with Learned Respiratory Dynamics.” *Proceedings of the International Symposium on Computer-based Medical Systems (CBMS)*. CBMS, 2024.
- Khatim, Nur Ahmad, Azmul Asmar Irfan, Amaliya Mata’ul Hayah, **Mansur Arief**. “Toward an Integrated Decision-Making Framework for Optimized Stroke Diagnosis with DSA and Treatment under Uncertainty.” *Proceedings of the 2024 6th International Conference on Intelligent Medicine and Image Processing (IMIP)*. IEEE, 2024.
- **Arief, Mansur**. Certifiable Evaluation for Safe Intelligent Autonomy. Carnegie Mellon University, 2023.
- Ding, Wenhao, Chejian Xu, **Mansur Arief**, Haohong Lin, Bo Li, Ding Zhao. "A Survey on Safety-Critical Driving Scenario Generation – A Methodological Perspective." *IEEE Transactions on Intelligent Transportation System*. IEEE, 2023
- Abdussyukur, Hafizh, Mahmud Dwi Sulistiyo, Ema Rachmawati, **Mansur Arief**. "Semantic Segmentation for Identifying Road Surface Damages Using Lightweight Encoder-Decoder Network." *Proceedings of the 2022 International Conference on Advanced Creative Networks and Intelligent Systems (ICACNIS)*. IEEE, 2022
- **Arief, Mansur**, Zhepeng Cen, Zhenyuan Liu, Zhiyuan Huang, Henry Lam, Bo Li, Ding Zhao. “Certifiable Evaluation for Autonomous Vehicle Perception Systems using Deep Importance Sampling (Deep IS).” *Proceedings of the 25th International Conference on Intelligent Transportation System (ITSC)*. IEEE, 2022.
- Nissa Amilia, Zulkifli Palinrungi, **Mansur Arief**, and Iwan Vanany. "Designing an Optimized Electric Vehicle Charging Station Infrastructure for Urban Area: A Case Study from Indonesia." *Proceedings of the 25th International Conference on Intelligent Transportation System (ITSC)*. IEEE, 2022.
- Ziyad, Muhammad, Kenrick Tjandra, Zulvah, Mushonnifun Faiz Sugihartanto, and **Mansur Arief**. "An Optimized and Safety-aware Maintenance Framework: A Case Study on Aircraft Engines." *Proceedings of the 25th International Conference on Intelligent Transportation System (ITSC)*. IEEE, 2022.

- **Arief, Mansur**, Zhiyuan Huang, Guru KS Kumar, Yuanlu Bai, Shengyi He, Wenhao Ding, Henry Lam, and Ding Zhao. "Deep Probabilistic Accelerated Evaluation: A Robust Certifiable Rare-Event Simulation Methodology for Safety-Critical Black-Box Autonomy." *Proceedings of the 24th International Conference on Artificial Intelligence and Statistics (AISTATS)*. PMLR, 2021.
- Arief, Hasan A., **Mansur Arief**, Guilin Zhang, Ulf Indahl, Håvard Tveite, and Ding Zhao. "SAnE: Smart annotation and evaluation tools for point cloud data". In *IEEE Access*. IEEE, 2020.
- Chen, Rui, **Mansur Arief**, Weiyang Zhang, and Ding Zhao. "How to Evaluate Self-Driving Testing Ground? A Quantitative Approach." *IEEE Transactions on Intelligent Transportation System*, DOI: 10.1109/ITITS.2020.2991757. IEEE, 2020.
- Huang, Zhiyuan, **Mansur Arief**, Henry Lam, and Ding Zhao. "Assessing Modeling Variability in Autonomous Vehicle Accelerated Evaluation." *Proceedings of the 22nd International Conference on Intelligent Transportation System (ITSC)*. IEEE, 2019.
- Guo, Yaohui, Vinay Varma Kalidindi, **Mansur Arief**, Wenshuo Wang, Jiacheng Zhu, Huei Peng, and Ding Zhao. "Modeling Multi-Vehicle Interaction Scenarios Using Gaussian Random Field." *Proceedings of the 22nd International Conference on Intelligent Transportation System (ITSC)*. IEEE, 2019.
- Liu, Zuxin, **Mansur Arief**, and Ding Zhao. "Where Should We Place LiDARs on the Autonomous Vehicle? An Optimal Design Approach." *Proceedings of the 2019 International Conference on Robotics and Automation (ICRA)*, pp. 2793-2799. IEEE, 2019.
- Arief, Hasan A., **Mansur Arief**, Manoj Bhat, Ulf Indahl, Håvard Tveite, and Ding Zhao. "Density-Adaptive Sampling for Heterogeneous Point Cloud Object Segmentation in Autonomous Vehicle Applications." *Proceedings of the 2019 IEEE Conference on Computer Vision and Pattern Recognition Workshops*, pp. 26-33. 2019.
- Huang, Zhiyuan, Yaohui Guo, **Mansur Arief**, Henry Lam, and Ding Zhao. "A versatile approach to evaluating and testing automated vehicles based on kernel methods." *Proceedings of the 2018 Annual American Control Conference (ACC)*, pp. 4796-4802. IEEE, 2018.
- Oktavian, Muhammad Rizki, Diana Febrita, and **Mansur Arief**. "Cogeneration Power-Desalination in Small Modular Reactors (SMRs) for Load Following in Indonesia." *Proceedings of the 4th International Conference on Science and Technology (ICST)*, pp. 1-6. IEEE, 2018.
- Huang, Zhiyuan, **Mansur Arief**, Henry Lam, and Ding Zhao. "Synthesis of Different Autonomous Vehicles Test Approaches." *Proceedings of the 21st International Conference on Intelligent Transportation Systems (ITSC)*, pp. 2000-2005. IEEE, 2018.
- **Arief, Mansur**, Peter Glynn, and Ding Zhao. "An Accelerated Approach to Safely and Efficiently Test Pre-Production Autonomous Vehicles on Public Streets." *Proceedings of the 21st International Conference on Intelligent Transportation Systems (ITSC)*, pp. 2006-2011. IEEE, 2018.
- Pujawan, Nyoman, **Mansur Arief**, Benny Tjahjono, and Duangpun Kritchanhai. "An integrated shipment planning and storage capacity decision under uncertainty: A simulation study." *International Journal of Physical Distribution & Logistics Management* 45(9), 2015.

PATENT

- Mike Timmerman, **Mansur Arief**, Jiachen Li, David Isele, Mykel Kochenderfer. "Importance sampling guided policy training." U.S. Patent Application 19/063748, filed February 26, 2025.
- **Arief, Mansur**, Ji Eun Kim, Shashank Shekhar, Mohammad Sadegh Norouzzadeh, and Ding Zhao. "Methods and systems of generating images utilizing machine learning and existing images with disentangled content and style encoding." U.S. Patent Application 17/949,517, filed April 4, 2024.

PRESENTATIONS & TALKS

- Guest Lecturer, *International Guest Lecture Series, ULBI, Bandung, Indonesia* 2025
- Invited Speaker, *Teach Talk 47, Indonesian Lecturer Association - Indonesian Diaspora Network United* 2025

■ Guest Lecturer, <i>Virginia Tech ECE 4524, Artificial Intelligence and Engineering Applications, Blacksburg, USA</i>	2025
■ Guest Lecturer, <i>UC San Diego MAE207 Safety for Autonomous, San Diego, USA</i>	2025
■ Session Chair, <i>American Geophysical Union (AGU) 2024, Washington DC, USA</i>	2024
■ Keynote Speaker, <i>2nd Int'l Conf. on Research in Engineering and Science Technology (IC-REST), Unhas, Gowa</i>	2024
■ Academic Speaker, <i>Indonesia Sustainability Forum (ISF), Jakarta</i>	2024
■ Invited Speaker, <i>Stanford Center for Earth Resources and Forecasting (SCERF), Stanford, USA</i>	2024
■ Invited Speaker, <i>Stanford Mineral-X, Stanford, USA</i>	2024
■ Invited Guest Lecture, <i>Net-Zero Heroes, Gawireia</i>	2024
■ Keynote Speaker, <i>8th Int'l Conf. on Management of Technology, Innovation, and Project (MOTIP 08)</i>	2024
■ Invited Public Guest Lecture, <i>Interdisciplinary School of Management Technology, ITS Surabaya, Virtual</i>	2023
■ <i>American Geophysical Union (AGU) 2023, San Francisco, USA</i>	2023
■ Invited Guest Lecture, <i>Industrial and Systems Engineering, ITS Surabaya</i>	2023
■ Invited Speaker, <i>Biweekly Meetings, IndoSTEELERS</i>	2023
■ Invited Speaker, <i>Teach Talk 41, Indonesian Lecturer Association - Indonesian Diaspora Network United</i>	2023
■ Invited Speaker, <i>International Webinar of Business and Industrial Application Division, BRAIN IPB</i>	2023
■ <i>25th International Conference on Intelligent Transportation Systems (ITSC), Macau, China</i>	2022
■ <i>Stanford Intelligent Systems Laboratory (SISL) Meeting</i>	2022
■ <i>24th International Conference on Artificial Intelligence and Statistics (AISTATS)</i>	2021
■ <i>SIAM Conference on Computational Science and Engineering (CSE21)</i>	2021
■ <i>INFORMS Annual Meeting</i>	2020
■ <i>2nd Annual Workshop of Safe AI Lab, Carnegie Mellon University, Pittsburgh, USA</i>	2020
■ <i>22nd International Conference on Intelligent Transportation Systems (ITSC), Auckland, NZ</i>	2019
■ <i>Workshop on Rare Events and Driving Scenario Modeling with Uber ATG, Pittsburgh, USA</i>	2019
■ <i>21st International Conference on Intelligent Transportation Systems (ITSC), Hawaii, USA</i>	2018
■ <i>Workshop on Safe AI, Carnegie Mellon University, Pittsburgh, USA</i>	2018

ACADEMIC SERVICES

■ Program Committee, <i>40th Annual AAAI Conference on Artificial Intelligence, Singapore,</i>	2026
■ Editor-In-Chief, <i>Operations and Supply Chain Management: An International Journal</i>	2025
■ Convener, <i>AGU Annual Meeting Session on Advancing Sustainable and Resilient Mineral Supplies</i>	2025
■ Guest Editor, <i>Environmental Research: Energy, Focus Issue - Critical Minerals and Subsurface Energy Resources</i>	2025
■ Program Committee, <i>Towards Reliable and Trustworthy Embodied AI in Everyday Scenarios Workshop, ICRA</i>	2025
■ Guest Editor, <i>Autonomous Systems in OSCM, Operations and Supply Chain Management: An International Journal</i>	2024
■ Primary Convener, <i>AGU Annual Meeting Session on Toward Sustainable Energy Transition Resources</i>	2024
■ Program Committee, <i>IEEE Int'l Conference on Automated Vehicle Validation Conference (IAVVC)</i>	2024
■ Program Committee, <i>Simulation and Modeling Methodologies, Technologies and Applications (SIMULTECH)</i>	2024
■ Program Committee, <i>International Conference on Intelligent Medicine and Image Processing (IMIP)</i>	2024
■ Reviewer, <i>Robotics and Automation Letter (RAL)</i>	2023
■ Reviewer, <i>Transportation Research Board Annual Meeting (TRBAM)</i>	2023
■ Reviewer, <i>Annals of Operations Research (ANOR)</i>	2023
■ Reviewer, <i>IEEE Transactions on Intelligent Vehicles (T-IV)</i>	2023

■ Reviewer, <i>Journal of Aerospace Information Systems (JAIS)</i>	2023
■ Reviewer, <i>Journal of Artificial Intelligence Research (JAIR)</i>	2023
■ Reviewer, <i>Operations and Supply Chain Management: An International Journal (OSCM)</i>	2023
■ Reviewer, <i>Heliyon</i>	2023
■ Program Committee, <i>Workshop on Formal Verification of Machine Learning (WVVML)</i>	2023
■ Program Committee, <i>Secure and Safe Autonomous Driving (SSAD) Workshop and Challenge</i>	2023
■ Reviewer, <i>International Design Engineering Technical Conferences (IDETC) Conference</i>	2023
■ Review Coordinator, <i>Design Theory and Methodology (DTM) Conference</i>	2023
■ Program Committee, <i>Trustworthy and Socially Responsible Machine Learning (TSRML) Workshop</i>	2022
■ Reviewer, <i>IEEE Transactions on Intelligent Transportation Systems (T-ITS)</i>	2022
■ Reviewer, <i>International Conference on Advanced Engineering Computing and Applications in Sciences</i>	2022
■ Reviewer, <i>IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)</i>	2022
■ Organizer, <i>Security and Safety in Machine Learning Systems, ICLR Workshop</i>	2021
■ Reviewer, <i>International Conference on Artificial Intelligence and Statistics (AISTATS)</i>	2021
■ Reviewer, <i>IEEE/RSJ International Conference on Robotics and Automation (ICRA)</i>	2019–2023
■ Reviewer, <i>International Conference on Intelligent Transportation Systems (ITSC)</i>	2018–2025
■ Technical Committee, <i>Operations and Supply Chain Management: An International Journal (OSCM)</i>	2014–2024

OTHER SERVICES

■ Founder, <i>AI Safety Indonesia</i>	2025
■ Co-Initiator, <i>MindSTEM.id</i>	2024
■ International Taskforce, <i>National Postdoctoral Association (NPA)</i>	2023
■ Founder, <i>Indonesian Scholars Thriving in Excellence in Education, Learning, and Research (IndoSTEELARS)</i>	2023
■ Volunteer, <i>AFS USA</i>	2022–2025
■ President, <i>Indonesian Students at CMU (PERMIAS CMU)</i>	2021
■ Vice President, <i>Indonesian Students at CMU (PERMIAS CMU)</i>	2020
■ Executive Board, <i>Indonesian Supply Chain and Logistics Institute, National Student Chapter</i>	2017
■ Executive Board, <i>INFORMS Student Chapter at University of Michigan</i>	2017
■ Volunteer, <i>AFS Bina Antar Budaya, Indonesia</i>	2008–2014